

SHETH MVLU COLLEGE

SUBJECT:- WSN

Aim:MANET Simulation with Directional Antenna

Simulate MANET using directional antennas and analyze improvements in spatial reuse and interference reduction.

CODE:-

```
set ns [new Simulator]

set tracefd [open directional_manet.tr w]
$ns trace-all $tracefd

set namfd [open directional_manet.nam w]
$ns namtrace-all-wireless $namfd 800 800

set topo [new Topography]
$topo load_flatgrid 800 800

create-god 5

set chan_1_ [new Channel/WirelessChannel]

$ns node-config \
    -adhocRouting AODV \
    -llType LL \
    -macType Mac/802_11 \
    -ifqType Queue/DropTail/PriQueue \
    -ifqLen 50 \
    -antType Antenna/OmniAntenna \
    -propType Propagation/TwoRayGround \
    -phyType Phy/WirelessPhy \
    -channel $chan_1_ \
    -topoInstance $topo \
    -agentTrace ON \
    -routerTrace ON \
    -macTrace ON \
    -movementTrace ON

for {set i 0} {$i < 5} {incr i} {
    set node_(i)[ns node]
}

$node_(0) set X_ 100
$node_(0) set Y_ 100
$node_(0) set Z_ 0

$node_(1) set X_ 250
```

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```
$node_(1) set Y_ 200  
$node_(1) set Z_ 0
```

```
$node_(2) set X_ 400  
$node_(2) set Y_ 300  
$node_(2) set Z_ 0
```

```
$node_(3) set X_ 550  
$node_(3) set Y_ 400  
$node_(3) set Z_ 0
```

```
$node_(4) set X_ 700  
$node_(4) set Y_ 500  
$node_(4) set Z_ 0
```

```
for {set i 0} {$i < 5} {incr i} {  
    $ns initial_node_pos node_$i 20  
}
```

```
set udp0 [new Agent/UDP]  
$ns attach-agent $node_(0) $udp0
```

```
set null0 [new Agent/Null]  
$ns attach-agent $node_(4) $null0
```

```
$ns connect $udp0 $null0
```

```
set cbr0 [new Application/Traffic/CBR]  
$cbr0 set packetSize_ 512  
$cbr0 set interval_ 0.1  
$cbr0 attach-agent $udp0
```

```
nsat1.0"node_(1) setdest 300 300 10"  
nsat2.0"node_(2) setdest 450 350 10"  
nsat3.0"node_(3) setdest 600 450 10"
```

```
nsat1.0"cbr0 start"  
nsat8.0"cbr0 stop"
```

```
proc finish {} {  
    global ns tracefd namfd
```

```
    $ns flush-trace
```

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```
close $tracefd  
close $namfd
```

```
puts "Simulation Completed"
```

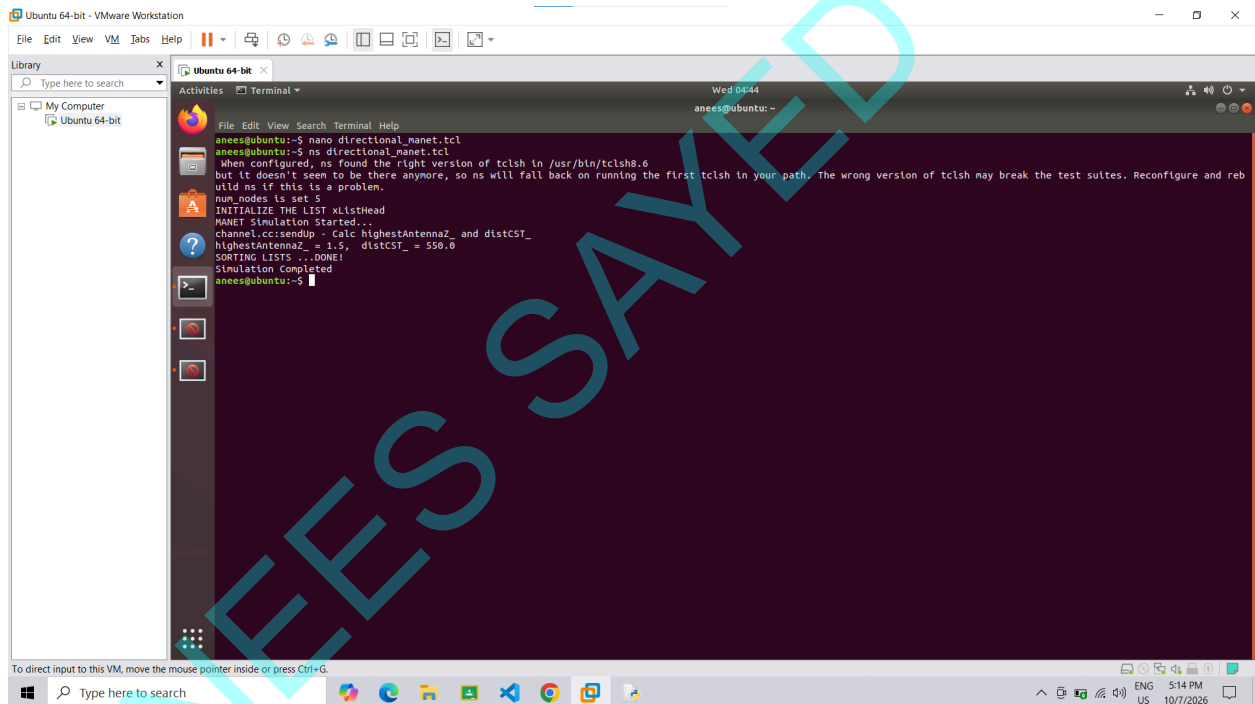
```
exec nam directional_manet.nam &  
exit 0
```

```
}
```

```
$ns at 10.0 "finish"
```

```
puts "MANET Simulation Started..."  
$ns run
```

OUTPUT:



```
anees@ubuntu:~$ nano directional_manet.tcl  
anees@ubuntu:~$ ns directional_manet.tcl  
When configured, ns found the right version of tcclsh in /usr/bin/tcclsh0.6  
but it doesn't seem to be there anymore, so ns will fall back on running the first tcclsh in your path. The wrong version of tcclsh may break the test suites. Reconfigure and reb  
uild ns if this is a problem.  
num_nodes is set 5  
INITIALIZE THE LIST xListHead  
MANET Simulation Started...  
channel.cc:sendUp - Calc highestAntennaZ_ and distcST_  
highestAntennaZ_ = 1.5, distcST_ = 550.0  
SORTING LISTS ...DONE!  
Simulation Completed  
anees@ubuntu:~$
```

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